Tutorial

Interpreting FAT Directory Entries

This tutorial introduces the process of manually reconstructing a file's starting cluster from a FAT directory entry.

1. Tutorial

Scenario

During a digital forensic examination of a FAT-formatted USB device, you locate the directory entry for a file called **notes.txt**, with a size of **24 KB**.

The directory entry contains the following information:

- **First cluster (MSB):** 0x0001
- **First cluster (LSB):** 0x0004

Task[1]

1. Combine the MSB and LSB values to determine the **starting cluster in hexadecimal**.
1. Convert this value into **decimal representation**.
2. Briefly explain what this cluster number represents.

Step 1: Understand the directory entry fields

In FAT file systems, the starting cluster number of a file is stored across two fields:

- The MSB contains the higher-order bits
- The LSB contains the lower-order bits

These two values must be combined to obtain the full starting cluster number.

Step 2: Combine the MSB and LSB values

Write the MSB followed by the LSB to form a single hexadecimal value:

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MSB: 0x0001
LSB: 0x0004


Combined hexadecimal value: 0x00010004
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[1] Data representation and number systems reference guide: https://www.computersciencecafe.com/112-number-systems-cie.html

Step 3: Convert the hexadecimal value to decimal

Convert the combined hexadecimal value into its decimal representation.

Show your working clearly.

(You may use a calculator for the conversion, but you must document the steps.)

Record your working:

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Step 4: Interpret the result

a. State the decimal cluster number you calculated.

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b. Explain what this value represents in the context of file storage.

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c. Does the file size (24 KB) fit within a single cluster, or would it require multiple clusters?

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d. How might fragmentation affect reconstruction?

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2. Exercise

Scenario

You are analysing a FAT-formatted memory card and attempt to reconstruct a deleted file called **image01.jpg**, with a size of **40 KB**.

The recovered directory entry shows:

- **First cluster (MSB):** 0x0000
- **First cluster (LSB):** 0x002A

Task

Follow the steps from the 1. Tutorial and work on the following task.

1. Calculate the starting cluster number in hexadecimal.

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2. Convert the value into decimal.

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3. State how many clusters the file would require if the cluster size is 4 KB.

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Additional Task - Optional

Consider **2. Exercise** and answer the following questions:

1. If the file is fragmented, why might knowing only the starting cluster be insufficient for full reconstruction?

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2. How might FAT allocation tables assist in this process, and what assumptions would you need to justify in a forensic report?

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